

223 Schema

The principal object in this schema is a *device* which is a tangible object designed to accomplish a specific task. Devices are connected to each other to form a *system*, which in turn may be grouped together to form other systems. Devices may have one or more *parts* which can provide additional details of the composition of the device.

- contains - links a system to one of its devices
- isContainedIn - links a device to the system that contains it

Containment relationships between systems are refined by these properties:

- hasSubsystem - links a system to one of its subsystems (subproperty of contains)
- isSubsystemOf - links a system to its supersystem (inverse hasSubsystem, subproperty of isContainedIn)

Containment relationships between systems and devices are refined by these properties:

- hasDevice - links a system to one of its subsystems (subproperty of contains)
- isDeviceOf - links a device to its system (inverse hasDevice, subproperty of isContainedIn)

Containment relationships between devices and their constituent parts are refined by these properties:

- hasPart - Relationship between a part and its parent part or device (subproperty of contains)
- isPartOf - inverse of hasPart (subproperty of isContainedIn)

System

A collection of connected devices or other systems.

Device

A tangible object designed to accomplish a specific task.

Devices may be connected to other devices. Connected devices interact in some ways.

- connectedTo - links a device or system to another device or system
- connectedFrom - links a device or system to another device or system from which it is connected to (inverse of connectedTo)

Devices have connection points that describe a way a device can be connected to another device.

- hasConnectionPoint - links a device or system to one of its connection points

- `connectedThrough` - links a device or system to a connection through which it is connected to other devices or systems by one of its connection points

Devices may have parts that contribute to its function.

Connection

Qualifies the `connectedTo` property between systems or devices.

Two or more connected systems or devices are connected through a connection.

- `connectsTo` - links a connection to one of the device or systems it connects
- `connectedAt` - links a connection to one of the connection points of the device to which is connects devices

ConnectionPoint

A connection point belongs to exactly one device. Any device connected through a connection is connected at one of its connection points to the connection.

- `connectsThrough` - links a connection point to the connection through which it connects its device or system
- `isConnectionPointOf` - links a connection point to its device

Part

A tangible object designed to be combined with other parts to refine the composition of a Device.

Property

An attribute, quality, or characteristic of a feature of interest.

- `hasProperty` - links a feature of interest to a property
- `isPropertyOf` - inverse of `hasProperty`
- `hasValue` - link from a Property to a Value

QuantifiableProperty

A property that can be quantified.

- `hasQuantityKind` - reference to `qudt:QuantityKind` (i.e., `quantitykind:Temperature`)
- `hasUnits` - reference to `qudt:Unit` (i.e., `unit:DEG_C`)

The `hasQuantityKind` property is required.

If the `hasUnits` property is provided then all of the Value instances (a) do not have a `hasUnits` property which implies that they are all the one associated with the property, or (b) if they do then the property is identical, e.g., both the Property and its associated Value instances reference `unit:DEG_C`.

If the **hasUnits** property is not provided, all of the **Value** instances must have a **hasUnits** property that is of the correct quantity kind, and the query engine is responsible for mapping the provided values to the expected values, e.g., **unit:DEG_C** is expected by **unit:DEG_F** is provided.

ObservableProperty

A subclass of **Property**, the value of the property is the result of a controller reading a sensor or calculating (see also **sosa:ObservableProperty**)

- **observesProperty** - link from an object/register reference to a property
- **isObservedBy** - link from the property to the object/register reference where it gets its value

ActuatableProperty

A subclass of **Property**, the value of the property is the result of a controller reading a sensor or calculating (see also **sosa:ActuatableProperty**)

- **actuatesProperty** - link from an object/register to a property
- **isActuatedBy** - link from the property to the object/register where it gets its value

Value

A representation of a value of a property (see also **qudt:Quantity**)

- **hasTimestamp** - datetime (see also **sosa:resultTime**)
- **hasSimpleValue** - literal value (see also **qudt:quantityValue**, **sosa:hasSimpleResult**)
- **hasUnits** - reference to **qudt:Unit** (i.e., **unit:DEG_C**)
- **isValueOf** - reference to a property, inverse of **hasValue**